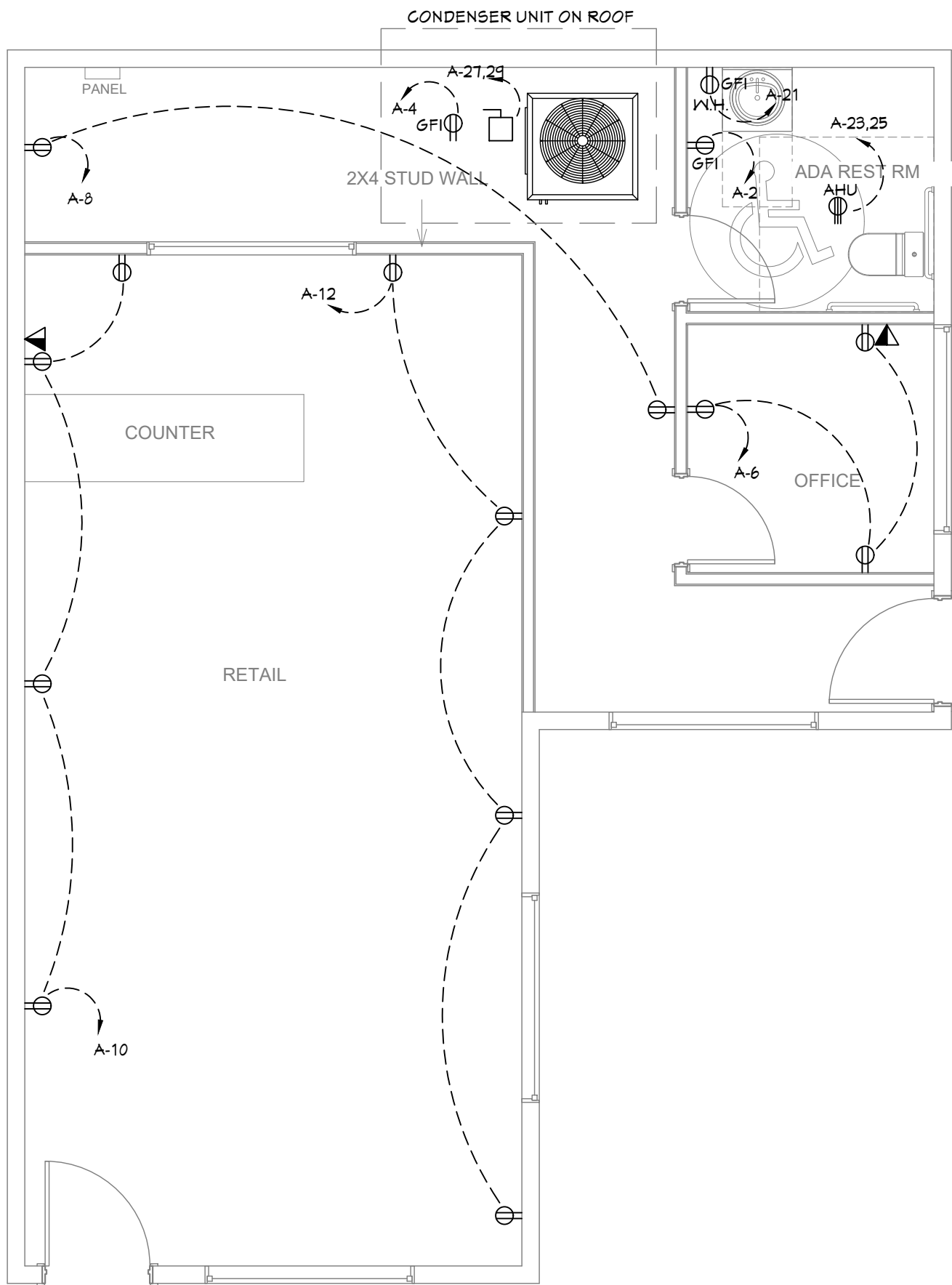
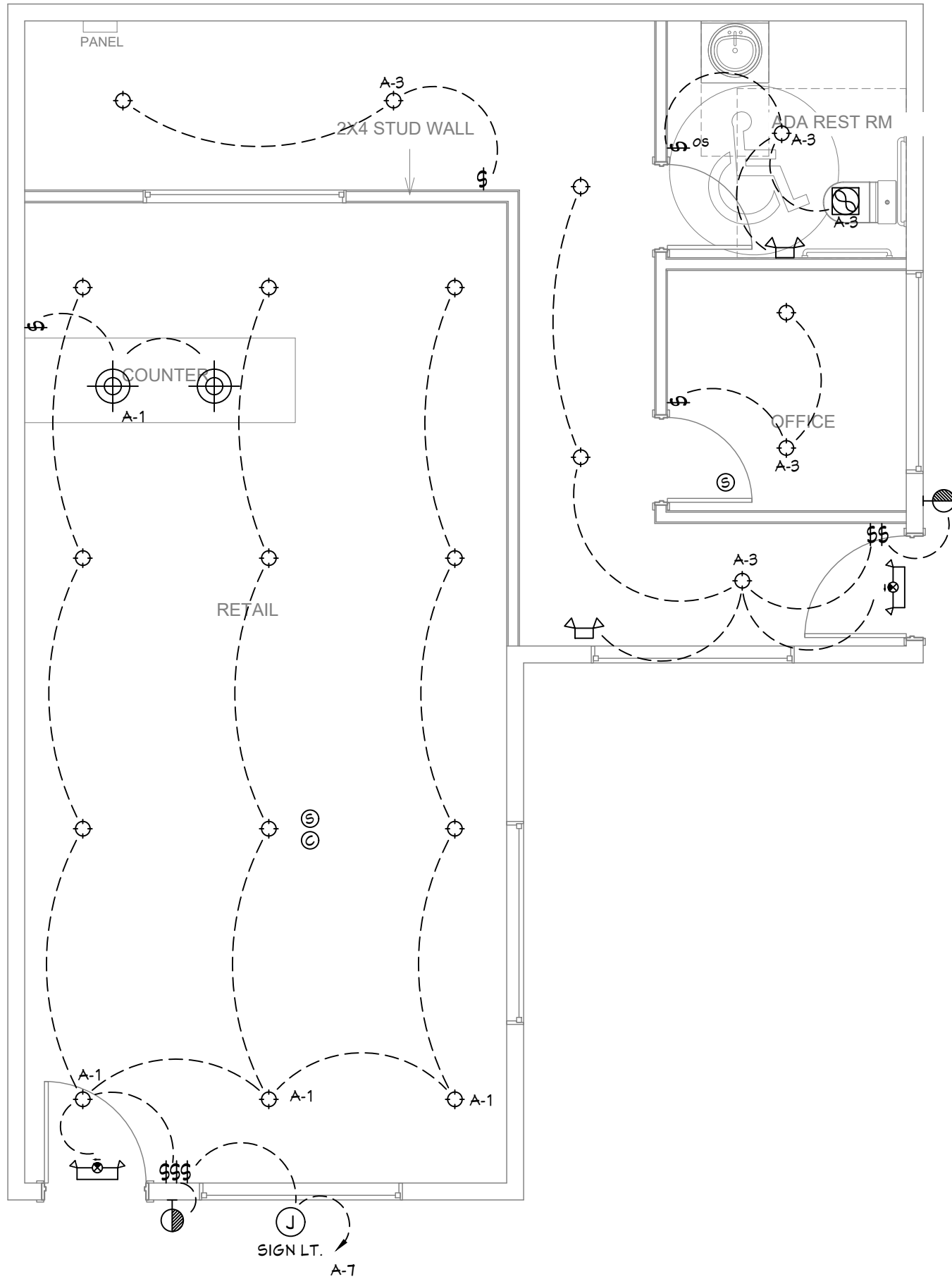




A-1



 **ELECTRICAL POWER PLAN**
SCALE: 1/4"=1'



 **ELECTRICAL LIGHTING PLAN**
SCALE: 1/4"=1'

LEGENDS	
	LED RECESSED DOWN LIGHT
	W/F LED RECESSED LIGHT
	CEILING LIGHT
	FAN WITH LIGHT KIT/CEIL. FAN
	W/F WALL LIGHT
	WALL LIGHT / VANITY LIGHT
	SENSOR SPOT LIGHT
	STRIP LIGHT
	PENDANT / CHANDLIER LIGHT
	DUPLEX REC.
	RECESSED DUPLEX REC.
	QUADRUPLUX REC.
	GROUND FAULT INTERRUPTER
	220V RECEPTACLE
	1/2 LIVE RECEPTACLE
	JUNCTION BOX
	EXHAUST FAN
	EXHAUST FAN WITH LIGHT
	SWITCH
	THREE WAY SWITCH
	FOUR WAY SWITCH
	FAN SWITCH
	OCCUPANCY SENSOR SW.
	PUSH BUTTON
	DATA + TV POINT
	MOTION SENSOR
	SMOKE DETECTOR
	CO DETECTOR
	DISCONNECT SWITCH
	EXIT LIGHT
	EXIT + EMERGENCY LIGHT
	EMERGENCY LIGHT

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PROPOSED RETAIL
W 19TH STREET HOUSTON,
TEXAS

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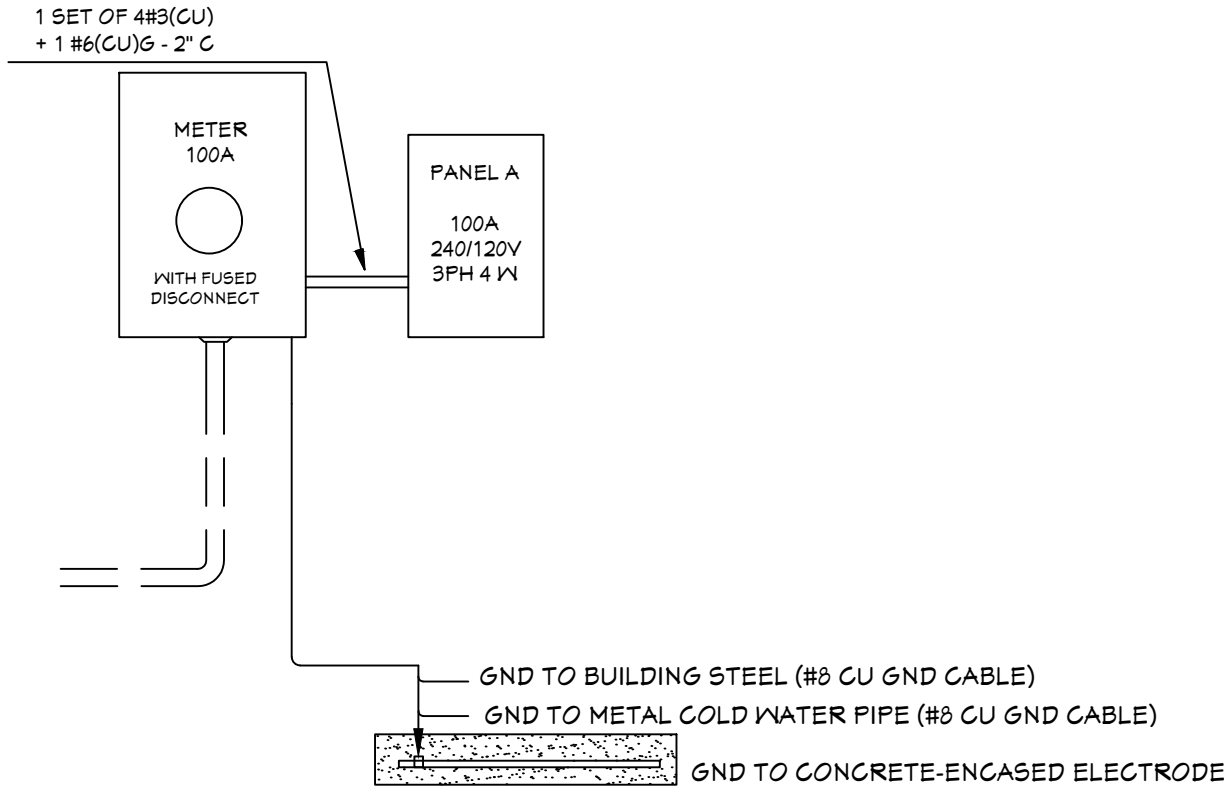
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ELECTRICAL
PLAN

E-I

PANEL SCHEDULE: PANEL A																
VOLTAGE: 208Y/120												PHASE: 3 , WIRES: 4				
DEVICE AMPS: 200						TYPE 1 (RECESSED)						AIC RATING: 10,000 A				
CKT #	ITEM	LOAD TYPE	LOAD (KVA)	TRIP/ POLE	WIRE SIZE/AWG	PH A	PH B	PH C	WIRE SIZE/AWG	TRIP/ POLE	LOAD (KVA)	LOAD TYPE	ITEM			CKT #
1	RETAIL LIGHTING	A	1.0	20/1	#12	1.2			#12	20/1	0.2	A	REST RM GFCI REC			2
3	REST RM+OFFICE+HALLWAY LIGHT	A	0.5	20/1	#12		0.7		#12	20/1	0.2	A	SERVICE GFCI REC @ ROOF			4
5	FIRE ALARM	A	0.2	20/1	#12			0.74	#12	20/1	0.5	A	OFFICE REC			6
7	SIGN LIGHT	A	0.5	20/1	#12	0.9			#12	20/1	0.4	A	HALLWAY+STORE REC			8
9	SPACE						0.7		#12	20/1	0.7	B	RETAIL REC			10
11	SPACE							0.72	#12	20/1	0.7	B	RETAIL REC			12
13	SPACE					0.0							SPACE			14
15	SPACE						0.0						SPACE			16
17	SPACE							0.00					SPACE			18
19	SPACE					0.0							SPACE			20
21	TANKLESS W.H.	E	3.5	30/1	#10		3.5						SPACE			22
23	AHU	D	0.5	20/2	#12			0.50					SPACE			
25		D	0.5		#12	0.5							SPACE			
27	2TON COND. UNIT	D	1.5	30/2	#10		1.5						SPACE			
29		D	1.5		#10			1.50					SPACE			
PANEL LOAD ANALYSIS																
LOAD TYPE	DESCRIPTION	CONN. KVA	DEMAND KVA	2020 NEC REFERENCE				LOAD TYPE	DESCRIPTION				CONN. KVA	DEMAND KVA	2020 NEC REFERENCE	
A	LIGHTING	3.46	2.8	ARTICLE 215.3				E	HEATING				3.50	3.50	ARTICLE 220.60	
B	RECEPTACLES	1.44	1.4	TABLE 220.44				F	LARGEST MOTOR				0.00	0.00	ARTICLE 440.7	
C	KITCHEN EQUIPMENT	0.00	0.0	TABLE 220.56				G	OTHER MOTORS				0.00	0.00	ARTICLE 440.7	
D	AIR-CONDITIONING	4.00	4.0	ARTICLE 440.32				H	OTHER LOADS				0.00	0.00	ARTICLE 220.14A	
PHASE A CONNECTED LOAD (KVA):		2.5	* DENOTES GFCI BREAKER					TOTAL CONNECTED LOAD (KVA):					12.4	KVA	34.4	AMP
PHASE B CONNECTED LOAD (KVA):		6.4						TOTAL DEMAND LOAD (KVA):					11.7	KVA	32.5	AMP
PHASE C CONNECTED LOAD (KVA):		3.5														



- NOTES:
- SERVICE GROUND PER NEC ARTICLE 250
 - THE EXACT A/C RATING OF ALL SERVICE EQUIPMENT AND PANELBOARDS SHALL BE COORDINATED WITH THE POWER COMPANY'S AVAILABLE FAULT CURRENT AND ADJUSTED ACCORDINGLY
 - ALL MULTI-CIRCUIT HOMERUNS SHALL BE PROTECTED WITH MULTI-POLE, SIMULTANEOUS-TRIP CIRCUIT BREAKERS PER N.E.C. 210.48
 - COMPLETE AFCI PROTECTION IS REQUIRED FOR ALL CIRCUITS, PER 2020 NEC.

POWER RISER

- NOTES:
- GFCI RECEPTACLES IN ALL BATHROOMS, NEAR SINKS, AND EXTERIOR RECEPTACLES.
 - ALL RECEPTACLES MUST BE AFCI PER NEC 2017 ARTICLE 210.12 & TEMPER RESISTANT PER NEC 2017 ARTICLE 406.12
 - ALL SMOKE DETECTORS SHALL BE HARDWIRED.
 - RESTROOM SWITCHES SHALL BE OCCUPANCY SENSOR.
 - ALL EMERGENCY LIGHTS SHALL BE BATTERY BACKUP (2 HOUR RUN TIME RATED).
 - PANEL BOARDS TO BE LABELED IN ACCORDANCE WITH NEC 408.4
 - GROUND ELECTRODE CONDUCTORS ARE INSTALLED IN ONE CONTINUOUS LENGTH WITHOUT SPLICES OR JOINTS
 - ALL CIRCUITS TO HAVE GROUND CONDUCTORS
 - ALL CIRCUITS ARE TO BE GROUND CONCEALED
 - EXTERIOR LIGHTING SHALL BE CONTAINED IN CUT-OFF TYPE LUMINARIES AND SHALL BE DIRECTED IN TOWARDS THE PROPERTY SO AS NOT TO REFLECT INTO ADJACENT RESIDENTIAL PROPERTIES
 - EXTERIOR LIGHTS ARE FULL CUT-OFF TYPE AND NO CHANGES SHALL BE MADE & TO BE ACT AS EMERGENCY LIGHTS.
 - THE BRANCH CIRCUIT FEEDING THE UNIT EMERGENCY LIGHTING SHALL BE THE SAME CIRCUIT AS THAT SERVING THE NORMAL LIGHTING IN THE AREA AND CONNECTED AHEAD OF ANY LOCAL SWITCHES
 - PROVIDE APPROPRIATE SEPARATION OF PANELS FROM ALL HAZARDS IN COMPLIANCE WITH NEC 2020.

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W 19TH STREET HOUSTON,
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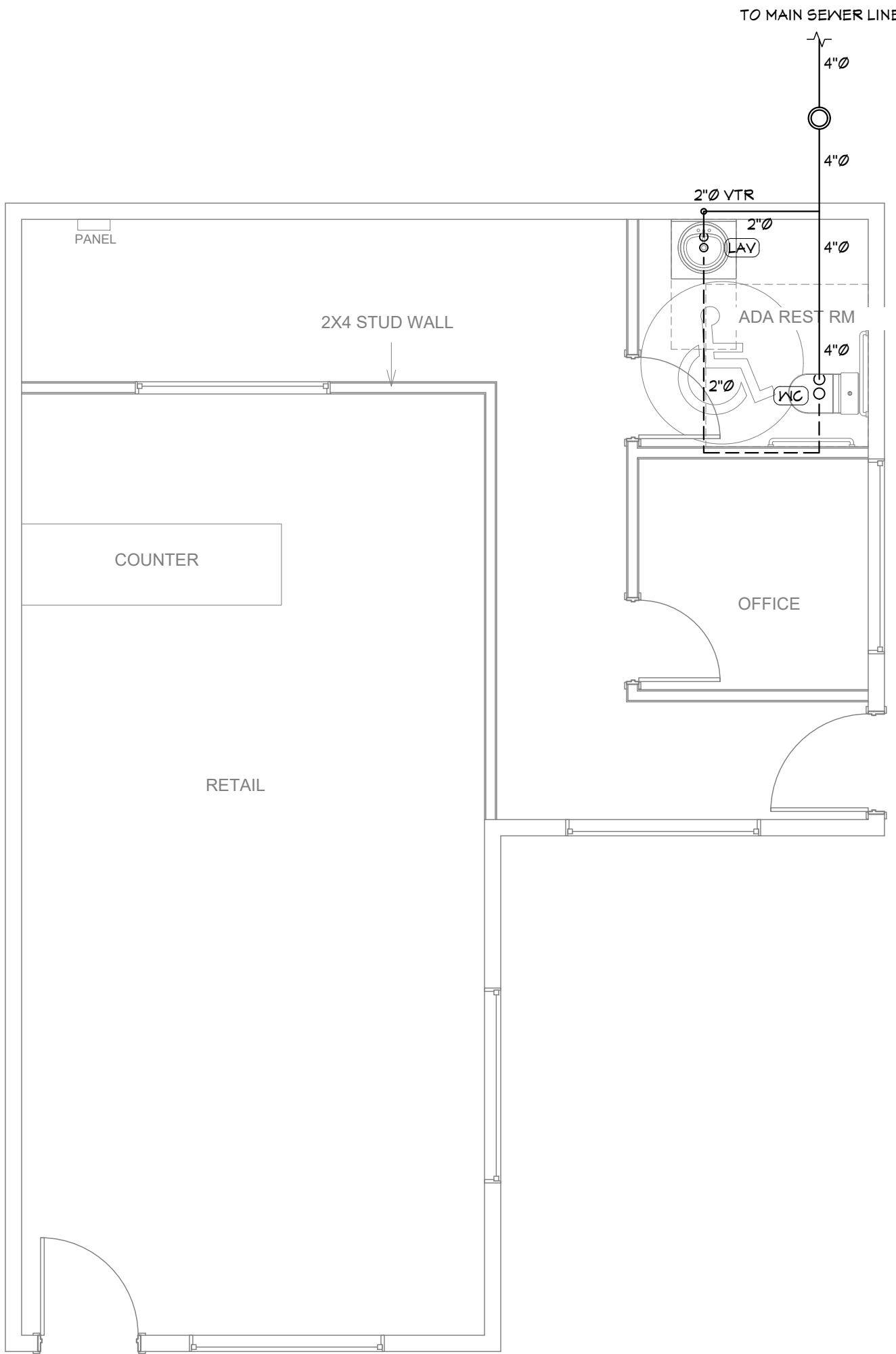
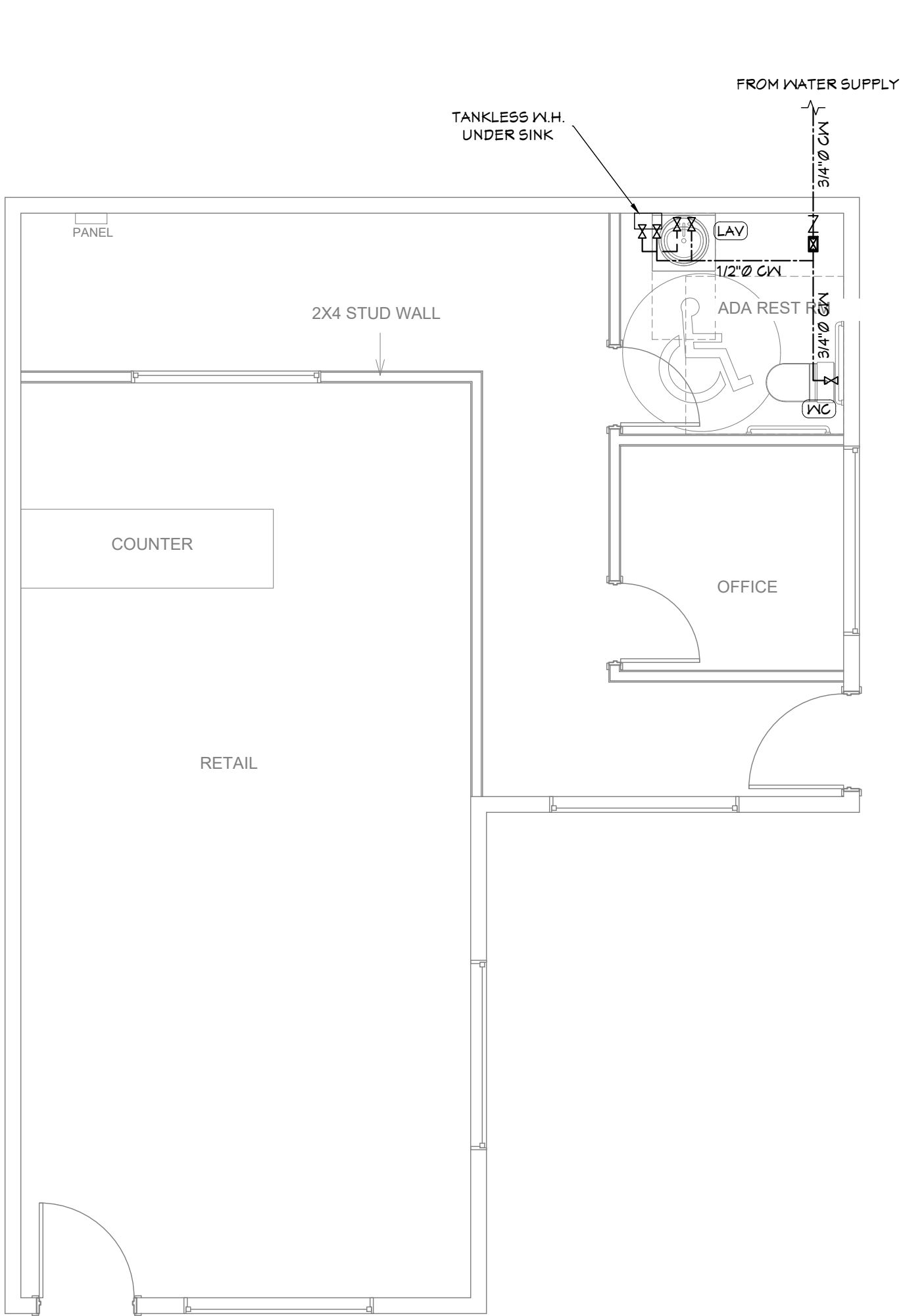
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ELECTRICAL
PANEL SCH.+SLD



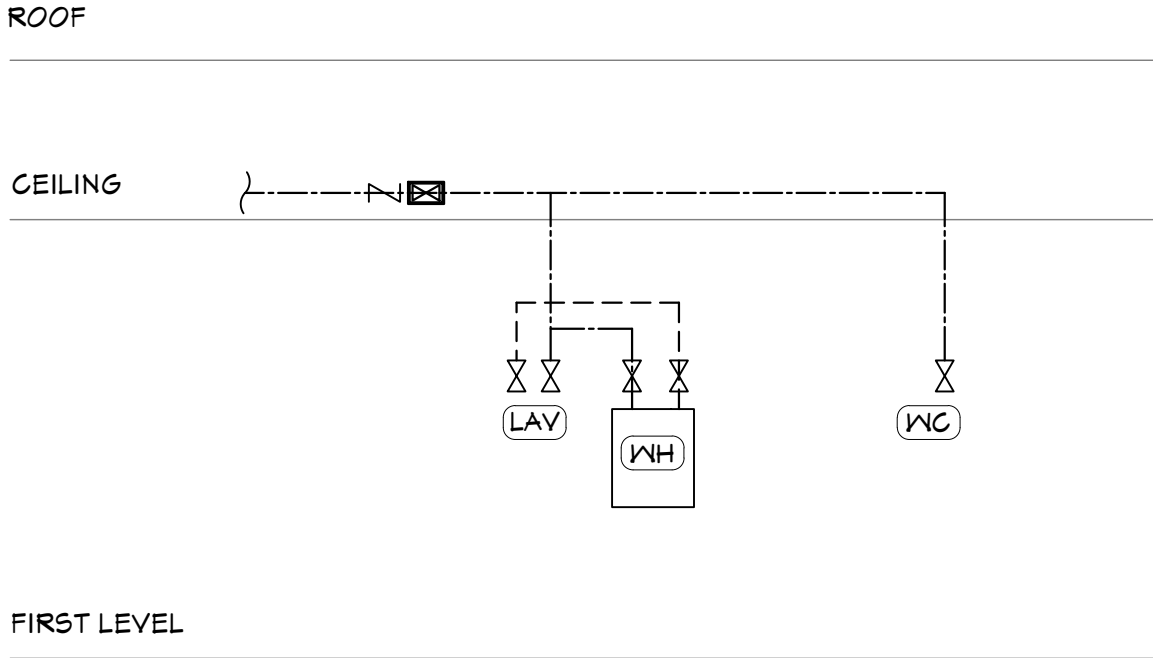
PLUMBING LEGEND & ABB.	
SYMBOL	DESCRIPTION
---	COLD WATER (CW)
---	HOT WATER (HW)
---	HOT WATER RETURN (HWR)
---	SANITARY WASTE LINE (SW)
---	VENT LINE (VP)
---	GAS
---	GAS
---	GAS LINE (GP)
---	GATE VALVE (G.V.)
---	BALL VALVE (B.V.)
---	CHECK VALVE (C.V.)
---	SHUT OFF VALVE (S.O.V.)
---	UNION
---	HOSE BIBB (H.B.)
---	BACKFLOW PREVENTER (B.F.P.)
---	FLOOR DRAIN (FD)
---	P-TRAP
---	FLOOR CLEANOUT
---	WALL CLEANOUT
---	CLEANOUT TO GRADE
---	PRESSURE RELIEF VALVE
---	VACUUM RELIEF VALVE
---	AUTOMATIC AIR VENT
---	A.A.V. AUTOMATIC AIR VENT
---	VENT THROUGH ROOF
---	WATER HEATER
---	LAVOTARY
---	BATH TUB
---	SHOWER
---	KITCHEN SINK
---	WASHER/DRYER
---	DISHWASHER
---	REFRIGERATOR
---	TO BELOW
---	TO ABOVE
---	FROM BELOW
---	FROM ABOVE
---	HIGH LEVEL
---	LOW LEVEL

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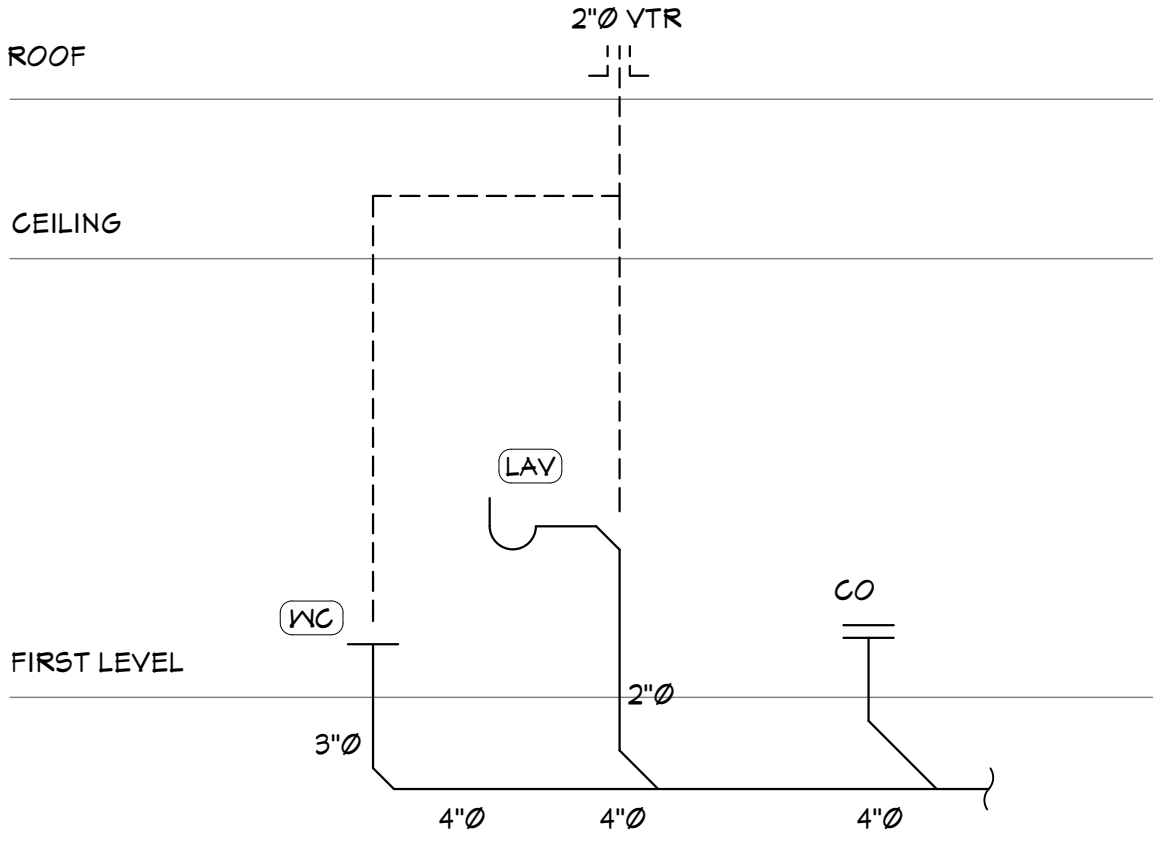
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PLUMBING PLAN		
P-I		



PLUMBING WATER RISER
NTS



PLUMBING SANITARY RISER
NTS

PLUMBING LEGEND & ABB.	
SYMBOL	DESCRIPTION
----	COLD WATER (CW)
----	HOT WATER (HW)
----	HOT WATER RETURN (HWR)
----	SANITARY WASTE LINE (WF)
----	VENT LINE (VP)
---GAS---GAS---	GAS LINE (GF)
⋈	GATE VALVE (G.V.)
⊕	BALL VALVE (B.V.)
⌞	CHECK VALVE (C.V.)
⊠	SHUT OFF VALVE (S.O.V.)
⋈	UNION
⊕	HOSE BIBB (H.B.)
⊠BFP	BACKFLOW PREVENTER (B.F.P.)
⊠	FLOOR DRAIN (FD)
⊠	P-TRAP
FCO	FLOOR CLEANOUT
WCO	WALL CLEANOUT
COTG	CLEANOUT TO GRADE
Ⓡ	PRESSURE RELIEF VALVE
Ⓥ	VACUUM RELIEF VALVE
ⓐ	AUTOMATIC AIR VENT
A.A.V.	AUTOMATIC AIR VENT
VTR	VENT THROUGH ROOF
W.H.	WATER HEATER
LAV.	LAVOTARY
BT	BATH TUB
SH	SHOWER
KS	KITCHEN SINK
WD	WASHER/DRYER
DW	DISHWASHER
REF	REFRIGERATOR
T/B	TO BELOW
T/A	TO ABOVE
F/B	FROM BELOW
F/A	FROM ABOVE
H/L	HIGH LEVEL
L/L	LOW LEVEL

PLUMBING SPECIFICATIONS:

INSTALL A COMPLETE, FUNCTIONAL PLUMBING SYSTEM COMPLYING WITH ALL ACCEPTABLE CODES, ORDINANCES, LAWS, AND REGULATIONS. DWV SYSTEMS AND WATER DISTRIBUTION PIPES TO COMPLY WITH LOCAL CODE.

OBTAIN PERMITS AND INSPECTIONS REQUIRED FOR THE INSTALLATION OF THIS WORK, AND PAY ALL CHARGES INCIDENT HERETO.

INSTALL ALL EQUIPMENT AND MATERIALS PER MANUFACTURER'S RECOMMENDATIONS.

WATER PIPING: PLASTIC WATER PIPE (CPVC & PEX) APPROVED. IF COPPER, USE TYPE 'L' HARD COPPER ABOVE GRADE, TYPE 'K' BURIED, LEAD-FREE SOLDER. ROUTE WATER PIPES SO IT WILL NOT BE EXPOSED TO FREEZING TEMPERATURES.

ALL ABOVE-GROUND WATER PIPING AND RELATED VALVES, FITTINGS, AND JOINTS SHALL BE INSULATED WITH ½" HEAVY DENSITY FIBERGLASS WITH FKS SELF-SEALING LAP JOINT JACKET. INSULATION MATERIALS SHALL HAVE COMPOSITE FIRE AND SMOKE HAZARD RATING AS TESTED BY UL PROCEDURE UL723, NOT EXCEEDING FLAME SPREAD RATING OF 25 AND SMOKE DEVELOPED RATING OF 50.

SOIL, WASTE, AND VENT PIPING SCHEDULE 40 DWV PVC PLASTIC PIPE WITH SOLVENT-WELDED JOINTS WHERE PERMITTED BY PLUMBING AND FIRE CODES; CAST IRON OTHERWISE.

PLUMBING NOTES:

ALL PLUMBING FIXTURES TO BE HIGH EFFICIENCY FIXTURES, PER LOCAL CODE. PLUMBER TO ENSURE VENT PIPES TERMINATE IN BUILDING EXTERIOR, WITH APPROPRIATE BIRD/PEST SCREENING PROTECTION. EVERY DRY VENT CONNECTED TO A HORIZONTAL DRAIN SHALL BE CONNECTED ABOVE THE CENTERLINE OF THE HORIZONTAL DRAIN PIPE. EVERY DRY VENT SHALL RISE TO A POINT NOT LESS THAN 6" ABOVE THE FLOOD LEVEL RIM OF THE HIGHEST TRAP OR TRAPPED FIXTURE BEING VENTED.

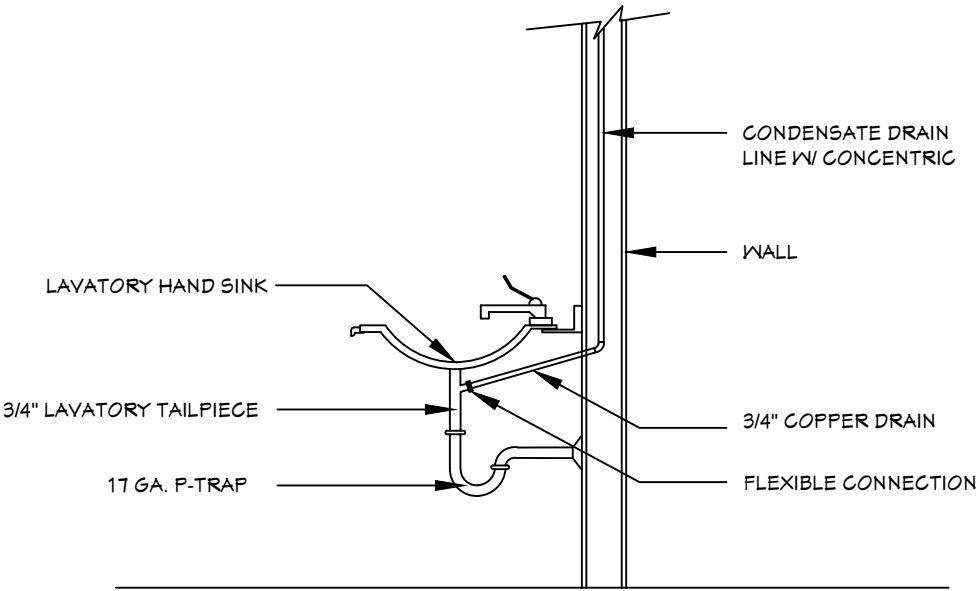
½" CONNECTIONS TO FIXTURES, TYP.

SLOPE OF HORIZONTAL PIPES SHOULD BE IN ACCORDANCE WITH IPC TABLE 704.1AS FOLLOWS:
PIPES 2½" OR LESS - ¼" PER FOOT, MIN.
PIPES 3 IN TO 6 IN - ⅛" PER FOOT, MIN.

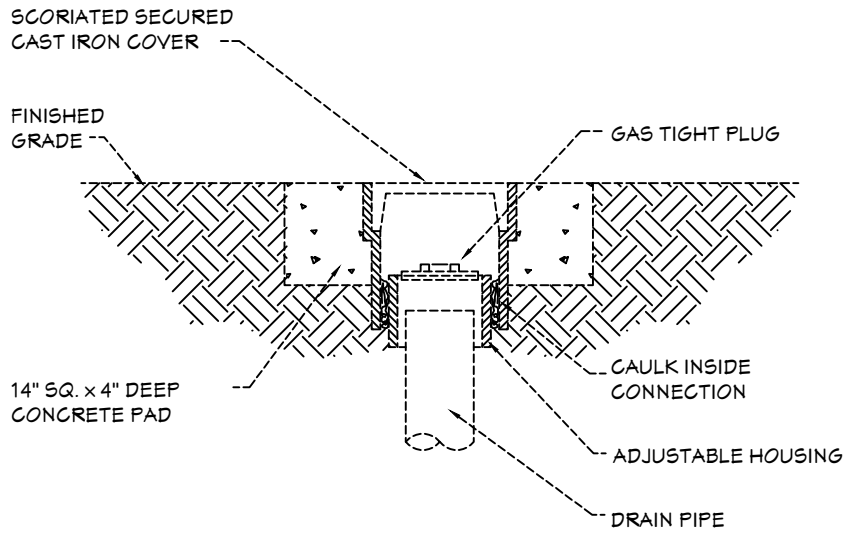
FIXTURE PIPE SIZES						
FIXTURE	MARK	CONNECTION				
		HW	CW	WASTE	TRAP	VENT
LAVATORY	LAV	1/2"	1/2"	2"	1 1/2"	2"
WATER CLOSET	WC	1/2"	1/2"	4"	INT.	2"
URINAL	UR	1/2"	1/2"	2"	2"	2"
SINK	S	1/2"	1/2"	2"	2"	2"
HOSE BIBB	HB		1/2"			
WATER FOUNTAIN	WF	1/2"	1/2"	2"	2"	2"
MOP SINK	W	1/2"	1/2"	3"	2"	2"
FLOOR DRAIN	FD			2"	2"	2"

ELECTRIC TANKLESS WATER HEATER				
MANUFACTURER	MODEL NO.	CAPACITY	DIMENSION	ELECTRI CAL
RHEEM	RTEX-04	0.3-1.5 GPM	8 x 10 x 3 inches	120/1/60

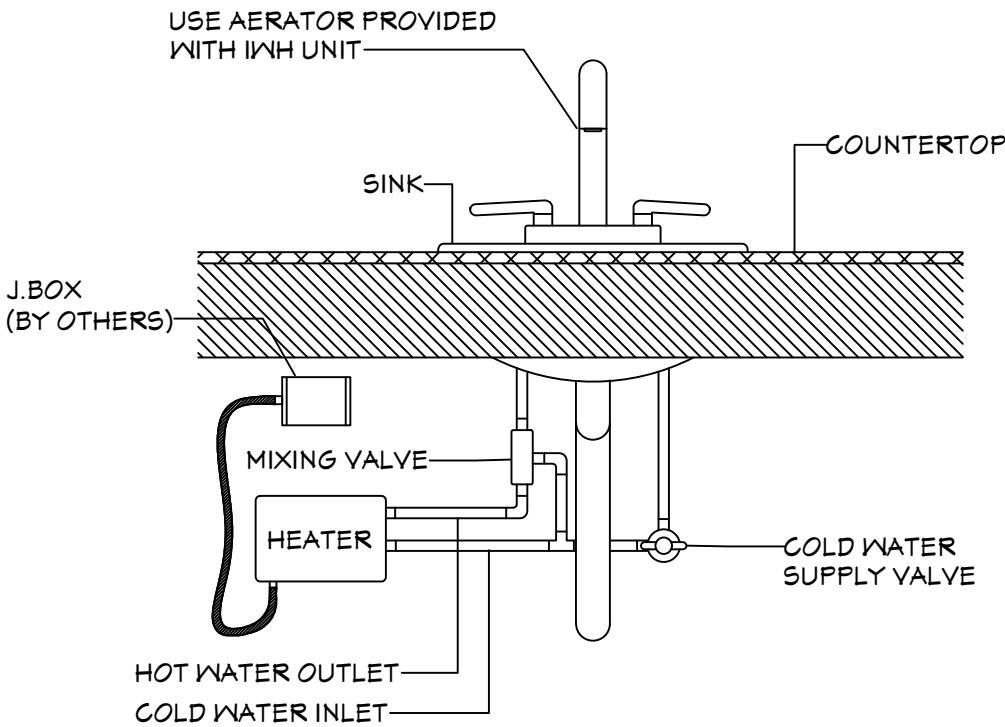
FIXTURE UNIT CALCULATION					
FIXTURE	QTY.	WATER FIXTURE UNIT	WASTE FIXTURE UNIT	TOTAL WATER FU	TOTAL WASTE FU
LAVATORY	1	2	2	2	2
WATER CLOSET	1	5	4	5	4
URINAL	0	2	2	0	0
MOP SINK	0	2	2	0	0
HOSE BIB	1	2.5	0	2.5	0
ADD H.B.	0	1	0	0	0
SINK	0	1.5	2	0	0
WATER FOUNTAIN	0	2	2	0	0
TOTAL				9.5	6
PRESSURE AVAILABLE				50-60 PSI	
MAX.EQUIVALENT LENGTH (FEET)				50	
SERVICE PIPE SIZE				3/4"	
DISTRIBUTION PIPE SIZE				3/4"	
MAIN DRAIN LINE				4"	



CONDENSATE DETAIL
NO SCALE



CLEANOUT TO GRADE
SCALE:NONE



INSTA-HOT WATER HEATER DETAIL
SCALE:NONE

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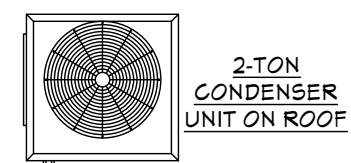
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PLUMBING
RISER+DETAILS



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MECHANICAL PLAN

M-1

2-TON HEAT PUMP SPLIT SYSTEM MECHANICAL SCHEDULE											
SYMBOL	MANUFACTURER*	MODEL*	CFM (MAX.)	OUTSIDE AIR	SEER	COOLING CAPACITY		HEATING CAPACITY		ELECTRICAL VOLT/PH/Hz	WEIGHT
COND. / AHU						NET MBH	NOM. MBH	NET MBH	NOM. MBH		
	GOODMAN	GSZ160241 AVFTC2B14	800	125	16	24	24	24	24	240/1/60	---

* OR APPROVED EQUAL
PROVIDE CONDENSATE DRAIN FOR ALL AHU UNITS

OUTDOOR AIR REQUIREMENTS						
SPACE	NET AREA, S.F. (B)	Rp(C) *	Rz(D) *	SPACE DENSITY *	SPACE OCCUPANCY (F)	O.A. REQ. (C x F + B x D) **
OFFICE	50	5	0.06	5 PER 1000 SQ FT	1	8
RETAIL	42	1.5	0.12	15 PER 1000 SQ FT	1	101.94
TOTAL						109.94
TOTAL OUTSIDE AIR PROVIDED ***						125
TOTAL AIR PROVIDED ***						800

* TABLE 6-1, ASHRAE 62.1-2010.
** VR = RP x P + RZ x A (PER ASHRAE 62.1-2010)
*** PROVIDED BY OUTSIDE AIR INTAKES

			EXHAUST FAN SCHEDULE					
SYMBOL	MANUFACTURER*	MODEL*	CFM	S.P. H2O	DRIVE	HP	ELECTRICAL VOLT/PH/Hz	COMMENTS
EXHAUST FAN (EF1)	BROAN	TBD	50	1/4	CENTRIFUGAL	0.135	120/1/60	SWITCH SEPARATELY, 4" DUCT, TERMINATE OUTSIDE WITH BACKDRAFT DAMPER.
EXHAUST FAN (EF2)	BROAN	TBD	100	1/4	CENTRIFUGAL	0.2	120/1/60	SWITCH SEPARATELY, 4" DUCT, TERMINATE OUTSIDE WITH BACKDRAFT DAMPER.

* OR APPROVED EQUAL

MECHANICAL SYMBOLS

	SUPPLY REGISTER, METAL, WHITE FINISH.
	SUPPLY REGISTER ON WALL, METAL, WHITE FINISH.
	RETURN GRILLE, METAL, WHITE FINISH.
	RETURN GRILLE ON WALL BOTTOM, METAL, WHITE FINISH.
	PROGRAMMABLE THERMOSTAT, HONEYWELL 8300 OR EQUAL, 54" A.F.F.
	EXHAUST FAN - SEE SPECIFICATIONS ON SCHEDULE BELOW. MECHANICALLY VENTED TO THE OUTSIDE.
	SUPPLY FLEXIBLE DUCTWORK (BUILDING STANDARD)
	RECTANGULAR DUCTWORK (BUILDING STANDARD)
	VOLUME DAMPER
	FIRE DAMPER
	PROGRAMMABLE THERMOSTAT
	DOOR UNDER-CUT 3/4" MIN.
	RETURN/EXHAUST AIR

ABBREVIATIONS

CD	CEILING DIFFUSER
COND	CONDENSATE
CFM	CUBIC FEET PER MINUTE
CLG	CEILING
CR	CEILING REGISTER
DB	DRY BULB
DN	DOWN
(E)	EXISTING
EF	EXHAUST FAN
ER	EXISTING RELOCATED
ESP	EXTERNAL STATIC PRESSURE
ETR	EXISTING TO RELOCATE
GAL	GALLONS
(N)	NEW
OBD	OPPOSED BLADE DAMPER
R	REMOVE
RR	REMOVE AND RELOCATE
SP	STATIC PRESSURE
T'STAT	THERMOSTAT
TYP	TYPICAL
VTO	VENT TO OUTSIDE
VD	VOLUME DAMPER
FL	FLOOR

AIR DEVICES SCHEDULE

CFM RANGE	LISTED SIZE	FACE SIZE	NC MAX.	MODEL	REMARKS
0-75	2 X 12	3-1/8 X 13-1/5	30	LIMA MODEL 875	REGISTER IN KICK PLATE UNDER SINK
0-50	6X6	7-1/2 X 7-1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
51-100	8X6	9-1/2 X 9-1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
101-150	12X6	13-1/2 X 7-1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
151-250	12X8	15-1/2 X 9-1/2	30	LIMA SERIES 600	WALL AND CEILING DIFFUSER
0-125	2-1/2 X 12	4-7/8 X 13-7/8	30	LIMA SERIES 667	FLOOR DIFFUSER
126-250	4 X 14	6-7/8 X 15-7/8	30	LIMA SERIES 667	FLOOR DIFFUSER
0-600	10 X 10	12 X 12	30	LIMA STEEL FILTERED GRILLE 60G	WALL/CEILING GRILLE
0-1500	22 X 22	24 X 24	30	LIMA STEEL FILTERED GRILLE 60G	WALL/CEILING GRILLE

NOTES:

- ① ALL CEILING DIFFUSER TO BE PROVIDED WITH:
A. SQUARE TO ROUND NECK TRANSITION.
B. OPPOSED BLADE DAMPER.
- ② SEE ARCH. REFLECTED CEILING PLAN FOR DIFFUSER AND REGISTER FOR EXACT LOCATION.

MECHANICAL NOTES AND SPECIFICATIONS

- THE CONTRACTOR SHALL PROVIDE A COMPLETE SYSTEM TO INCLUDE ALL LABOR MATERIALS, TOOLS, AND EQUIPMENT FOR A COMPLETE AND FUNCTIONAL SYSTEM INCLUDING ALL NECESSARY APPURTENANCES CUSTOMARILY INCLUDED IF NOT NECESSARILY CALLED OUT.
- DEVIATION FROM MATERIALS, METHODS, AND PROCEDURES SET FORTH HEREIN MUST BE APPROVED IN WRITING. APPROVAL WILL NOT BE GIVEN UNLESS THE PROPOSED SYSTEM IS EQUAL IN PERFORMANCE, DURABILITY, LONGEVITY, AND RELIABILITY TO THAT SPECIFIED.
- ALL WORK SHALL BE PERFORMED IN COMPLIANCE WITH ALL LOCAL, STATE, AND FEDERAL CODES, LAWS, AND ORDINANCES.
- THE CONTRACTOR SHALL VISIT THE JOB SITE AND THOROUGHLY FAMILIARIZE HIMSELF WITH ALL DETAILS OF THE WORK AND THE CONDITIONS, AND SHALL VERIFY DIMENSIONS AND CLEARANCES AND BE ASSURED THAT THE EQUIPMENT PURCHASED WILL FIT INTO THE AVAILABLE SPACE.
- SUBMISSION OF A BID OR PROPOSAL SHALL BE CONSTRUED AS EVIDENCE THAT THE CONTRACTOR HAS FAMILIARIZED HIMSELF WITH THE PLANS, SPECIFICATIONS AND BUILDING SITE. CLAIMS MADE SUBSEQUENT TO BIDS FOR MATERIALS AND/OR LABOR DUE TO DIFFICULTIES ENCOUNTERED WILL NOT BE RECOGNIZED UNLESS DIFFICULTIES COULD NOT HAVE BEEN FORESEEN THOUGH PROPER EXAMINATION HAD BEEN MADE.
- THE CONTRACTOR SHALL REVIEW STRUCTURAL DRAWINGS BEFORE FABRICATING OR INSTALLING DUCTWORK OR EQUIPMENT TO AVOID ANY CONFLICTS.
- ALL HARD DUCTWORK SHALL BE FABRICATED OF STEEL OF THICKNESS AND GAUGES TO CONFORM TO SMACNA DUCT CONSTRUCTION STANDARDS (RECTANGULAR OR EQUIVALENT ROUND).
- ALL DUCTS SHALL BE AIRTIGHT, FREE OF LEAKS, AND SHALL BE INSPECTED FOR LEAKS PRIOR TO INSTALLATION OF FINISHED CEILING SYSTEM.
- DUCTWORK, DAMPERS, LOUVERS, DIFFUSERS, AND OTHER DISTRIBUTION EQUIPMENT AND MATERIALS SHALL CONFORM TO THE FOLLOWING:

A. ASHRAE/ANSI

B. SBCCI/ASME

C. SMACNA

D. NFPA

E. AMCA STANDARD HANDBOX 99

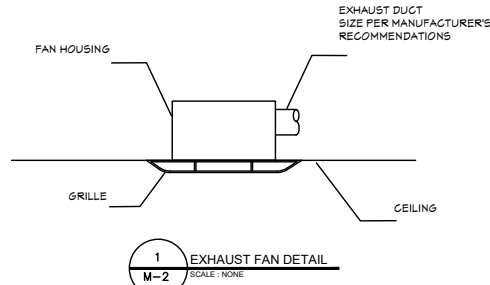
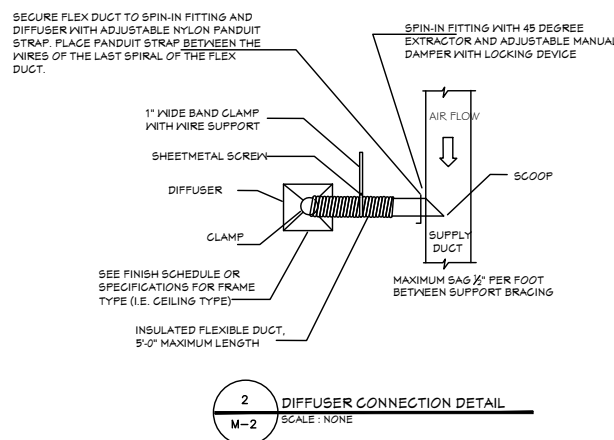
F. AIR DIFFUSION COUNCIL TEST CODE 1062R3

G. ANY LOCAL CODES NOT COVERED ABOVE
- ALL DUCT SIZES ARE CLEAR NET INSIDE DIMENSIONS.
- ALL DUCTWORK SHALL BE CONTINUOUS WELDED, LEAK-FREE, STEEL CONSTRUCTION CONFORMING TO SMACNA CODES. ACCESS PANELS INSTALLED NEAR ELBOWS TO FACILITATE CLEANING.
- ALL MAKE-UP AIR DUCTWORK SHALL BE INSULATED WITH 1-1/2", 3/4# DENSITY FOIL-BACKED EXTERNAL INSULATION.
- MANUAL DAMPERS SHALL BE INCLUDED IN OUTSIDE AIR INTAKE DUCTS TO REGULATE OUTSIDE AIR TO EQUIPMENT.
- ALL INSULATING MATERIAL SHALL HAVE A COMPOSITE FLAME SPREAD INDEX OF 25, AND SMOKE DEVELOPED INDEX OF 50 (ASTM E-84, NFPA 255 & UL 723).
- APPLICABLE FIRE DAMPERS SHALL BE RUSKIN MODEL D-IBD2 STYLE "B" OR APPROVED EQUAL WITH A 1-1/2 HOUR RATING.
- BALANCE ALL AIR SYSTEMS TO PRODUCE THE VOLUMES AND QUANTITIES SHOWN ON THE DRAWINGS OR SPECIFIED.
- THE CONTRACTOR SHALL VERIFY ALL ELECTRICAL CHARACTERISTICS WITH ELECTRICAL DRAWINGS BEFORE PURCHASING EQUIPMENT.

NOTES:

- THE HVAC SYSTEM FOR THIS RETAIL STORE SHALL CONSIST OF 2-TON HEAT PUMP SPLIT SYSTEM, WITH THE CONDENSER OUTSIDE ON A 4" THICK LEVEL PAD AT ROOF, AND THE HORIZONTAL FLOW AIR-HANDLER UNIT ON A PLATFORM IN THE BATH CLG AS SHOWN.
- RETURN AIR FILTER WILL BE AT THE AHU.
- INSTALL FUSIBLE LINK FIRE DAMPER ON CLOTHES DRYER DUCT.
- HVAC DUCTWORK SHALL BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH AIR CONDITIONING CONTRACTORS OF AMERICA (ACCA) DUCT SIZING CALCULATION BASED ON A FRICTION LOSS OF 0.1. THE FOLLOWING CHART IS BASED ON THE ACCA MANUAL:

PIPE DIAMETER	FLEX PIPE CFM
6"	0 - 100
8"	101 - 250
10"	251 - 400
12"	401 - 700
14"	701 - 1,000
16"	1,000 - 1,500



PROPOSED RETAIL
W 19TH STREET HOUSTON,
TEXAS

CONSULTANTS

OWNER

MARK DATE DESCRIPTION

PAGE SIZE:

ARCH FULL BLEED C (24.00 X 18.00 INCHES)

SCALE:

NOTE ON PLAN

DRAWN BY:

CHK'D BY:

SHEET TITLE

MECHANICAL
DETAILS

M-2